

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 11: Areas Related to Circles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Sector and segment of a circle

1. In a circle of radius 21 cm, an arc subtends 60° at the centre. Find (i) the length of the arc (ii) the area of the minor segment cut off by the chord of this arc. **CBSE 2024, 5 marks**

SOLUTION

$$(i) \text{ Length of arc} = \frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{6} \times 132 = 22 \text{ cm}$$

$$(ii) \text{ Area of sector} = \frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = 231 \text{ cm}^2$$

The triangle formed by the two radii and the chord has an angle of 60° between equal sides, so it is equilateral with side 21 cm. Its area = $\frac{\sqrt{3}}{4} \times 21^2 = \frac{441\sqrt{3}}{4} \text{ cm}^2$

$$\text{Area of minor segment} = \left(231 - \frac{441\sqrt{3}}{4} \right) \text{ cm}^2 \approx 231 - 190.95 = 40.05 \text{ cm}^2 \text{ (taking } \sqrt{3} = 1.732 \text{)}$$

$$\text{Answer: (i) 22 cm (ii) } \left(231 - \frac{441\sqrt{3}}{4} \right) \text{ cm}^2 \approx 40.05 \text{ cm}^2$$

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